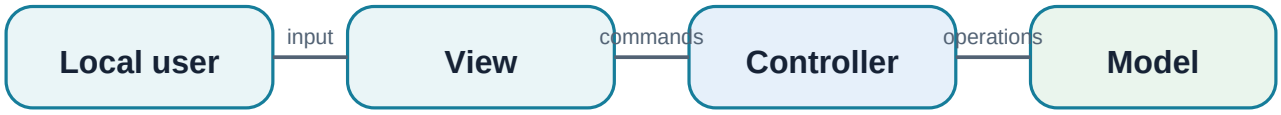
Design Document

COMP3211 Software Engineering | Fall 2026

Visual design diagrams

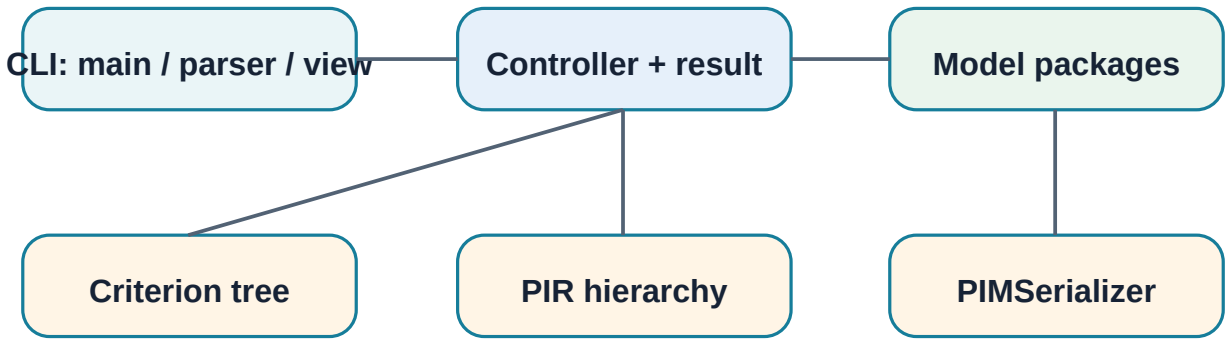
The architecture, component relationships, and search sequence are rendered below for the PDF deliverable.

Architecture

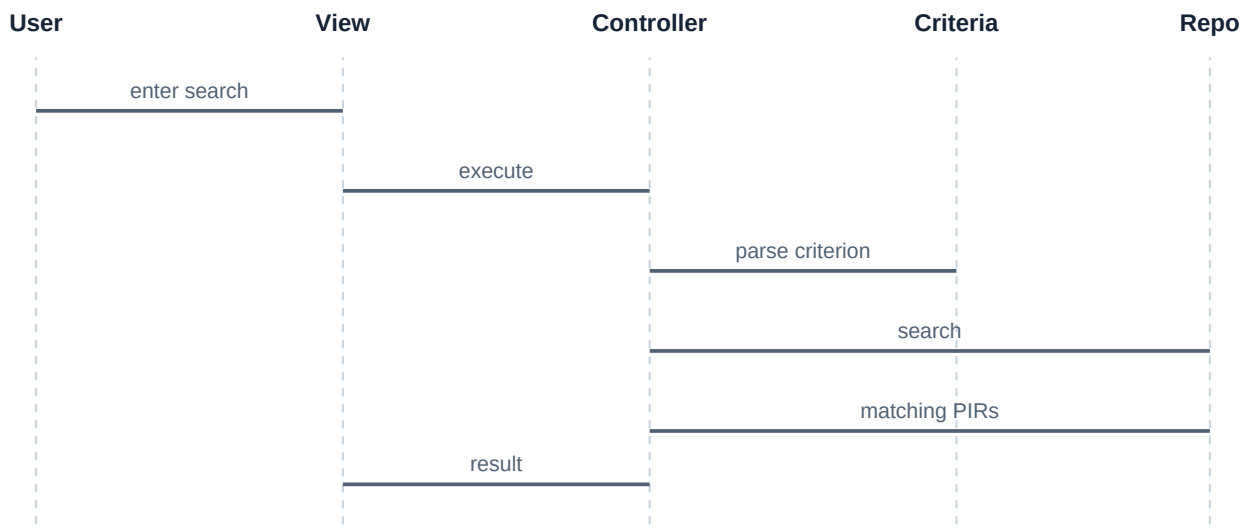


PIRs, criteria, repository, and .pim serializer

Main component relationships



Example use: search then update



Design Document

Status: Final submission baseline Version: 1.0 System: Hong Kong Student Second-hand PIM

1 Design goals

The design turns the Stage 1 requirements into a small, testable command-line application. It keeps the domain model independent from terminal input and output, gives every PIR type a consistent interface, and makes search and persistence replaceable components.

The design targets Python 3.11 or later and Python standard libraries only. The final developer manual will record the exact Python version and documented development platform.

2 Architectural pattern

The system uses Model-View-Controller (MVC).

- **Model** owns PIR data, validation, search evaluation, repository operations, and `.pim` serialization.
- **View** owns prompts and human-readable terminal output.
- **Controller** receives parsed commands, invokes model operations, and converts domain errors into operation results.

MVC is suitable because the model must be unit tested without simulating terminal input. It also keeps command syntax changes in the CLI layer and protects the model from presentation details.

The model never imports the CLI package. The controller is the only component that coordinates the repository, criterion parser, and serializer for a user command.

3 Main component relationships

Component responsibilities

Component	Responsibility	Depends on
`main.py`	Create objects and run the command loop	CLI components
`CommandParser`	Convert one input line into a typed command	`shlex`, command errors
`Controller`	Execute create, update, search, print, delete, save, load, help, and exit	Model components, result type
`View`	Display prompts, success output, records, and errors	Result type
`PIRRepository`	Allocate IDs, store records, update records, search, and delete	PIR hierarchy, criteria
PIR hierarchy	Validate and represent record-specific fields	Python standard library
`CriterionParser`	Parse the search grammar into a criterion tree	Criterion classes, PIR field names
Criterion classes	Evaluate predicates and logical operators	PIR interface
`PIMSerializer`	Validate `.pim` paths and round-trip records	`json`, `pathlib`, PIR factory
Domain errors	Give callers stable error categories	`Exception`

4 Domain class design

Shared PIR contract

Method	Signature	Return	Exceptions	
`allowed_fields`	`allowed_fields() -> frozenset[str]`	Fields valid for the concrete type	None	
`get_field`	`get_field(name: str) -> object \`	None`	Field value, or `None` when the record has no such field	`ValidationError` for a unknown field name

Method	Signature	Return	Exceptions	
update	update(changes: Mapping[str, object]) -> None	None	ValidationError if a field is unknown, empty, or has the wrong type	
to_dict	to_dict() -> dict[str, Any]	Serializable record data	None after a valid object exists	
format_details	format_details() -> str	Stable human-readable details	None	

Constructors validate IDs, required strings, and time values. Date and time fields are stored as `datetime` objects in memory and serialized as `YYYY-MM-DDTHH:MM` strings.

Concrete PIR fields

Class	Constructor	type_name	Allowed fields
PlainTextPIR	(id: int, content: str)	text	content
TaskPIR	(id: int, description: str, deadline: datetime)	task	description, deadline
EventPIR	(id: int, description: str, start: datetime, alarm: datetime)	event	description, start, alarm
ContactPIR	(id: int, name: str, address: str, mobile: str)	contact	name, address, mobile

5 Repository design

`PIRRepository` stores records in a private dictionary keyed by ID. It returns records in ascending ID order for `all` and `search` so output is deterministic.

Method	Signature	Return	Exceptions
create_plain_text	(content: str) -> PlainTextPIR	New record	ValidationError
create_task	(description: str, deadline: datetime) -> TaskPIR	New record	ValidationError
create_event	(description: str, start: datetime, alarm: datetime) -> EventPIR	New record	ValidationError
create_contact	(name: str, address: str, mobile: str) -> ContactPIR	New record	ValidationError
get	(record_id: int) -> PIR	Existing record	PIRNotFoundError
update	(record_id: int, changes: Mapping[str, object]) -> PIR	Updated record	PIRNotFoundError, ValidationError
search	(criterion: Criterion) -> list[PIR]	Matching records	None after criterion validation
all	() -> list[PIR]	All records in ID order	None
delete	(record_id: int) -> None	None	PIRNotFoundError
replace	(records: Sequence[PIR]) -> None	None	ValidationError for duplicate invalid IDs

`update` validates all changes before mutating the record. If one field is invalid, no field from the update is applied.

6 Search design

The criterion parser converts the Stage 1 grammar into a tree of `Criterion` objects.

Class or method	Signature	Return	Exceptions
`Criterion.evaluate`	`(record: PIR) -> bool`	Whether the record matches	`ValidationError` only for an in constructed criterion
`PredicateCriterion`	`(field: str, operator: str, value: object)`	Criterion object	`ValidationError` for unsuppor field/operator/value
`CriterionParser.parse`	`(expression: str) -> Criterion`	Parsed criterion tree	`CriterionParseError`

The parser enforces precedence `!`, then `&&`, then `||`. Text `contains` comparisons are case-insensitive. A missing field on a concrete PIR evaluates to `False`; an unknown field in the expression is a parse error.

7 Persistence design

`PIMSerializer` is the only class that reads or writes `.pim` files.

Method	Signature	Return	Exceptions
`save`	`save(records: Sequence[PIR], path: Path) -> None`	None	`PersistenceError`, `ValidationError`
`load`	`load(path: Path) -> list[PIR]`	Validated records	`PersistenceError`, `ValidationError`

The Stage 2 schema is:

```
{
  "schema_version": 1,
  "records": [
    {"id": 1, "type": "text", "content": "..."},
    {"id": 2, "type": "task", "description": "...", "deadline": "2026-11-20T18:00"}
  ]
}
```

The serializer rejects a non-`.pim` path, a missing required top-level key, an unsupported schema version, duplicate IDs, unknown types, invalid fields, and invalid time strings. It writes through a temporary file and replaces the target only after serialization succeeds.

8 CLI design

Class or method	Signature	Return	Exceptions
`CommandParser.parse`	`(line: str) -> Command`	Parsed command	`CommandSyntaxError`
`Controller.execute`	`(command: Command) -> OperationResult`	Result for the view	Catches domain errors and return `error` status
`View.render`	`(result: OperationResult) -> None`	None	None for valid result objects
`main.run`	`() -> None`	None	Handles end-of-input and clean ex

`Controller.execute` maps `PIMError` subclasses into `OperationResult(status="error", ...)`. The CLI loop therefore continues after recoverable input errors and never exposes a Python traceback during normal use.

9 Example use: search then update

The sequence below satisfies the design-document example-use requirement.

10 Design-to-requirements mapping

Requirement area	Design element
US1-US5	PIR hierarchy and repository create methods
US6	`PIR.update` and repository atomic update
US7	`CriterionParser` and criterion tree
US8-US9	Repository `all`, `get`, and `delete`; View formatting

Requirement area	Design element
US10-US11	`PIMSerializer` and repository `replace`
Model unit tests	Model has no CLI dependency
Standard library	Python classes, `datetime`, `json`, `pathlib`, `shlex`, and `typing`